Two-Block 4+4→8 Circuit Formation: Stationary Theory versus Training

Result in one sentence. Neither protocol forms the 4+4→8 circuit in any of the theory's 34 H_C cells: both fit the $d8$ target directly in 1k steps, before the parents are accessible, and stay in the direct phase H_D in all 52 cells that keep it.

Experiment. The setting of the $d4/d8$ theory: exact eight-bit cube, parents $\chi_L = x_1x_2x_3x_4$, $\chi_R = x_5x_6x_7x_8$, target $\chi_8 = \chi_L\chi_R$ ($\gamma_4 = \gamma_8 = 2$), the bias-free two-block residual ReLU network of the $d2/d4$ runs ($N = 1024$, float64; both readouts read the final stream h_2), and $\lambda_E = \lambda_2 = \lambda_p = \lambda_8 = \lambda_0$, $\lambda_1 = (\lambda_1/\lambda_2)\lambda_0$ on a 9×8 grid ($\lambda_0 = 10^{-4}$–1, $\lambda_1/\lambda_2 = 0.1$–100), seed 0, both protocols. SGD with momentum 0.9 and rate 10^{-3} for every parameter group, standard initialization (variance 1/fan-in for every matrix, readouts included; no μP), exactly 250k steps, no noise or early stopping. Parent-warm cells start from E, V_1, W_{d4} fitted to the parents read from h_1 (10k steps, same optimizer; $R_4 = 0.097$).

Measurements. Phase labels, f_C, φ_{out}, φ_{sq} and f_C^{sq} are those of the Langevin report with χ_L, χ_R as parents and χ_8 as target; $R_4 = Q_4/Q_1^{(1)}$ (gate 5/32). New (Fig. 4): (a) the block-2 share of the parent amplitude (exact, as $h_0 = Ex$ holds only singletons); (b) what block 2's $d8$ producers read: the $|w_{8,n}a_n|$-weighted share of their pre-activation power on the singletons (direct route D_2), the parents (C_{44}), the χ_8 mode already in h_1 (a relay of the block-1 route D_1), or degree-2 leakage; (c) $J/\min(F_C, F_D, F_0)$, the final objective over the theory's minimal free energy ($F_0 = 3$: zero network). A cell is *settled* when J changed by $< 1\%$ over the last 50k steps (29 of the 57 non-suppressed cells per protocol).

Seed 0, 9×8 grid	Parent-warm	Joint	Lower objective
Phase agreement with theory	45.8%	45.8%	45.8%
Gate $H_C \Leftrightarrow R_4 > 5/32$ (active cells)	73.1%	76.9%	76.9%
Mean $\lvert f_C - f_C^{\text{theory}}\rvert$	0.379	0.380	0.379
Cells in H_C / H_D / parents only / suppressed (theory: 34/18/0/20)	0/52/5/15	0/52/5/15	0/52/5/15
Cells won (parent-warm / joint)			43/29
R_4 when the target is first fitted: median (max)	0.093 (0.160)	0.067 (0.140)	
$J/\min(F_C, F_D, F_0)$ in settled cells	0.33–0.93	0.33–0.93	
Protocols select different phases	0/72 cells; final J differs by $\leq 1.8\%$ (median 0.04%)		

Observations. (i) *No composition where it is predicted.* In all 34 theory-H_C cells both protocols end in H_D with $f_C \leq 0.05$ (theory 0.66–0.88) and $\varphi_{\text{out}} < 10^{-5}$ (theory 1); outside $\lambda_0 = 0.3$ the 4+4→8 share never exceeds 0.06 along any trajectory. The protocols agree in every cell, so the lower-objective row repeats them; pretraining stops at $R_4 = 0.097 < 5/32$ and does not help. (ii) *The target is fitted before the parents are accessible.* Every cell that ever fits χ_8 ($A_8 \geq 0.5$) does so by the first snapshot (1k steps), at $R_4 = 0.001$–0.16 (below $5/32 = 0.156$ in all but one cell). R_4 crosses the gate only later (1k–230k steps) and only for $\lambda_1/\lambda_2 \leq 1$, $\lambda_0 \geq 10^{-3}$ (the 14 parent-warm and 12 joint H_D cells where the gate disagrees), and the direct routes are not replaced. (iii) *Block 2 relays or reads raw bits instead of composing* (Fig. 4). Its $d8$ producers read the singletons ($\geq 97\%$, route D_2) for $\lambda_1/\lambda_2 \geq 5$, and for $\lambda_1/\lambda_2 \leq 1$ (settled cells) the χ_8 mode that block 1 has already built (77–99%): block 2 amplifies D_1, a route outside the theory's two families; $\lambda_1/\lambda_2 = 2$ mixes the two. Parents supply $\leq 9\%$ and degree-2 leakage $\leq 8\%$ of these reads, so the 2+2+2+2→8 route the theory warns about is not used. (iv) *The parents are partly built in block 2*, since their readouts read h_2: 41–64% of the joint runs' parent amplitude for $\lambda_1/\lambda_2 \leq 1$ and 78–99% for $\lambda_1/\lambda_2 \geq 10$ (parent-warm 15–58% and 52–99%), whereas the theory charges both parents to block 1 (cost $\propto \lambda_1$). Block-2 parents cannot feed a 4+4→8 unit. (v) *Training is cheaper than the closure allows.* In settled cells $J = (0.33$–$0.93)\min(F_C, F_D, F_0)$, lowest at large λ_1/λ_2 where the theory pays λ_1 for the parents; in the 15 settled theory-H_C cells the H_D networks reach $J = (0.66$–$0.86)\,F_C$, so the free-energy comparison that places H_C does not bound what training reaches. At $\lambda_0 = 0.1$, $\lambda_1/\lambda_2 \geq 5$ (theory: suppressed) training keeps the parents (80–99% from block 2) and part of the target ($A_8 = 0.30$–0.46, $J = 2.35$–$2.61 < F_0$). The only compositional block 2, at $(\lambda_0, \lambda_1/\lambda_2) = (0.3, 0.1)$, is also predicted suppressed: $\varphi_{\text{out}} = 1.00$, $f_C = 0.52$, but $A_8 = 0.22$ ($J = 2.78$); its neighbour $(0.3, 0.3)$ stalls at $J = 3.33 > F_0$. (vi) *Weak decay is not stationary.* Decay alone shrinks the random initial hidden weights ($\|V\|^2 \approx N$) only by $e^{-\eta\lambda t/(1-\beta)} = e^{-0.25}$ at $\lambda = 10^{-4}$ ($e^{-2.5}$ at 10^{-3}) in 250k steps. The 28 unsettled cells (all with $\lambda_0 \leq 10^{-3}$ or $\lambda_1 \leq 10^{-3}$, plus the marginal $(3 \times 10^{-3}, 2)$) still lose 1–51% of J per 50k steps, with J up to 54× the theory's minimum; they include 19 of the 34 theory-H_C cells and do not test the stationary theory. (vii) As for $d2/d4$, readout-free and output-weighted shares agree (Fig. 2), and the final MSE is set by λ_0, identically for both protocols (Fig. 3).

Limits. One seed and one optimizer setting. With the target fitted within 1k steps and block 2 free to relay D_1 or build the parents, this run does not isolate the theory's two-route competition; that needs parents read from h_1, χ_8 and degree-2 leakage projected out of h_1, parents pretrained to $R_4 > 5/32$ before the target is switched on, and μP or a larger rate so that weak-decay cells become stationary.

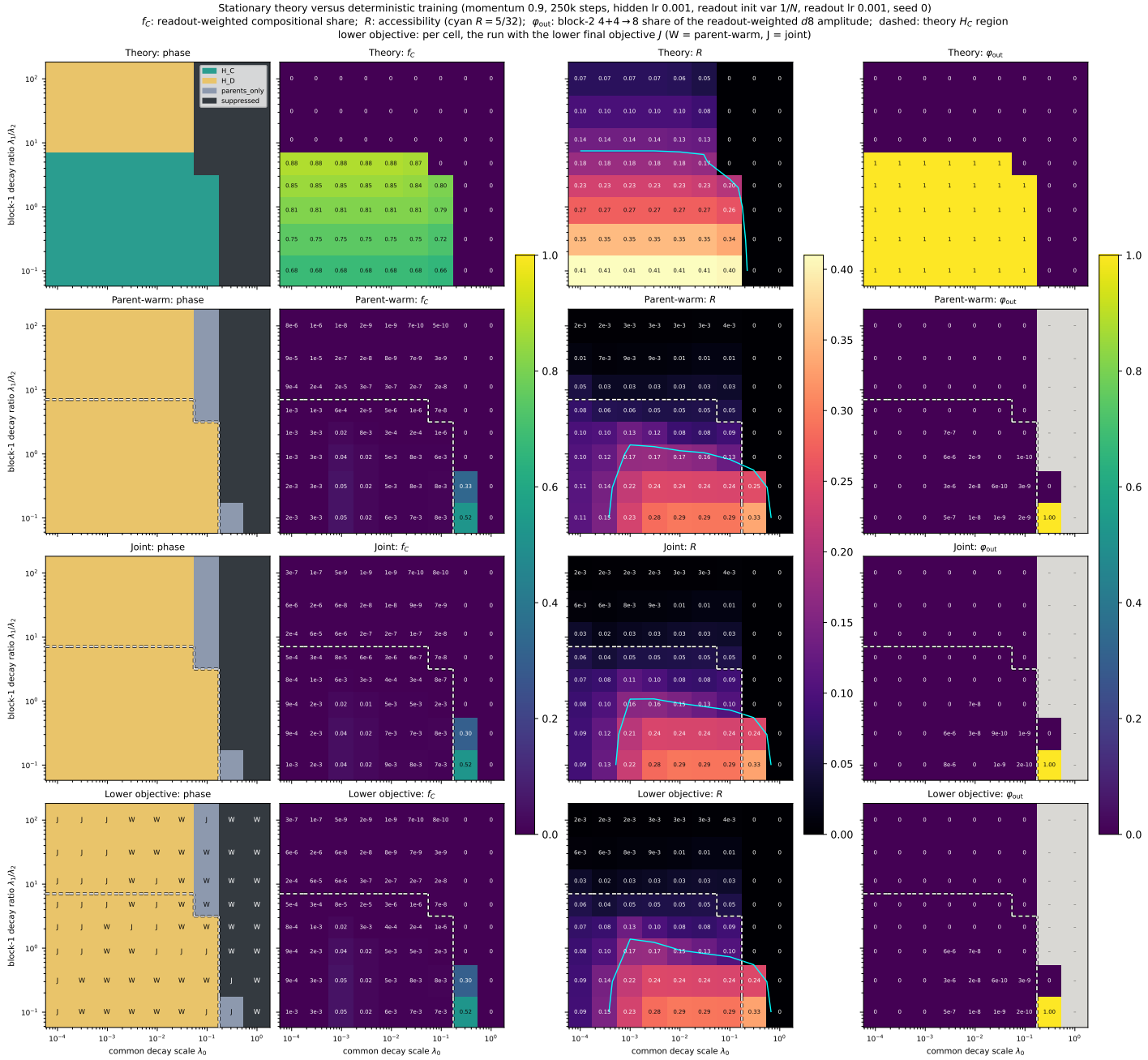


Figure 1: **Theory, parent-warm, joint, and the lower-objective run per cell (rows, top to bottom).** In the bottom row each cell shows whichever of the two runs ended with the lower final objective J (W = parent-warm, J = joint). Columns: phase, readout-weighted compositional share f_C, accessibility R_4 (cyan: $R_4 = 5/32$), and the block-2 4+4→8 share φ_{out} of the readout-weighted $d8$ amplitude. Dashed: theory H_C region. Grey: no target mode produced.

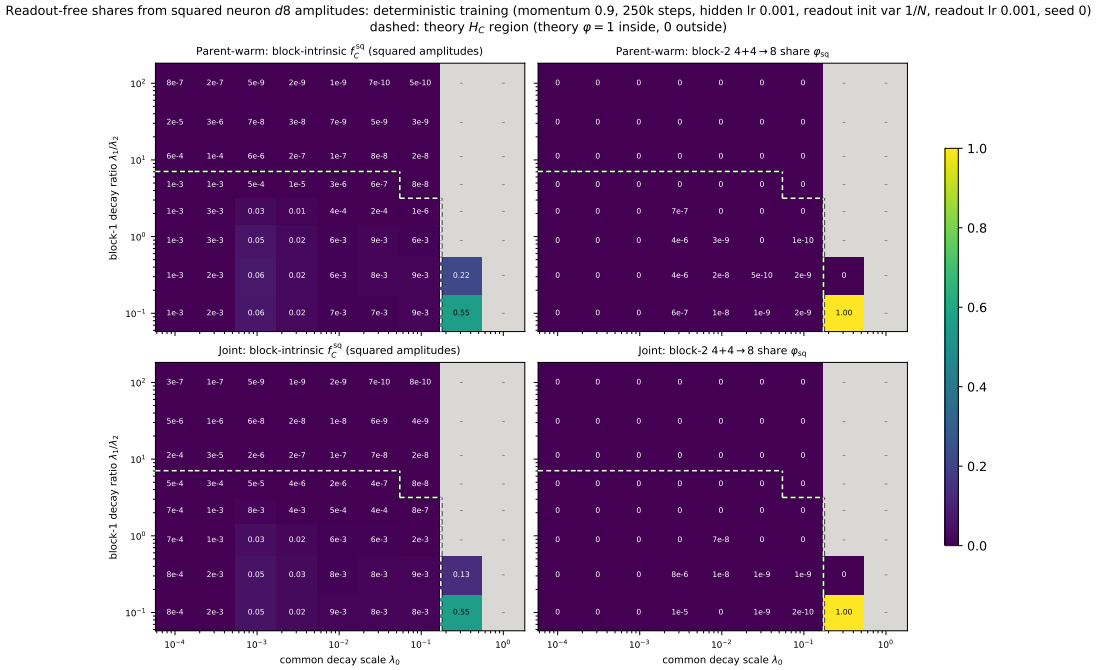


Figure 2: **Readout-free shares** from squared neuron $d8$ amplitudes: f_C^{sq} (left) and φ_{sq} (right); compare with columns 2 and 4 of Fig. 1.

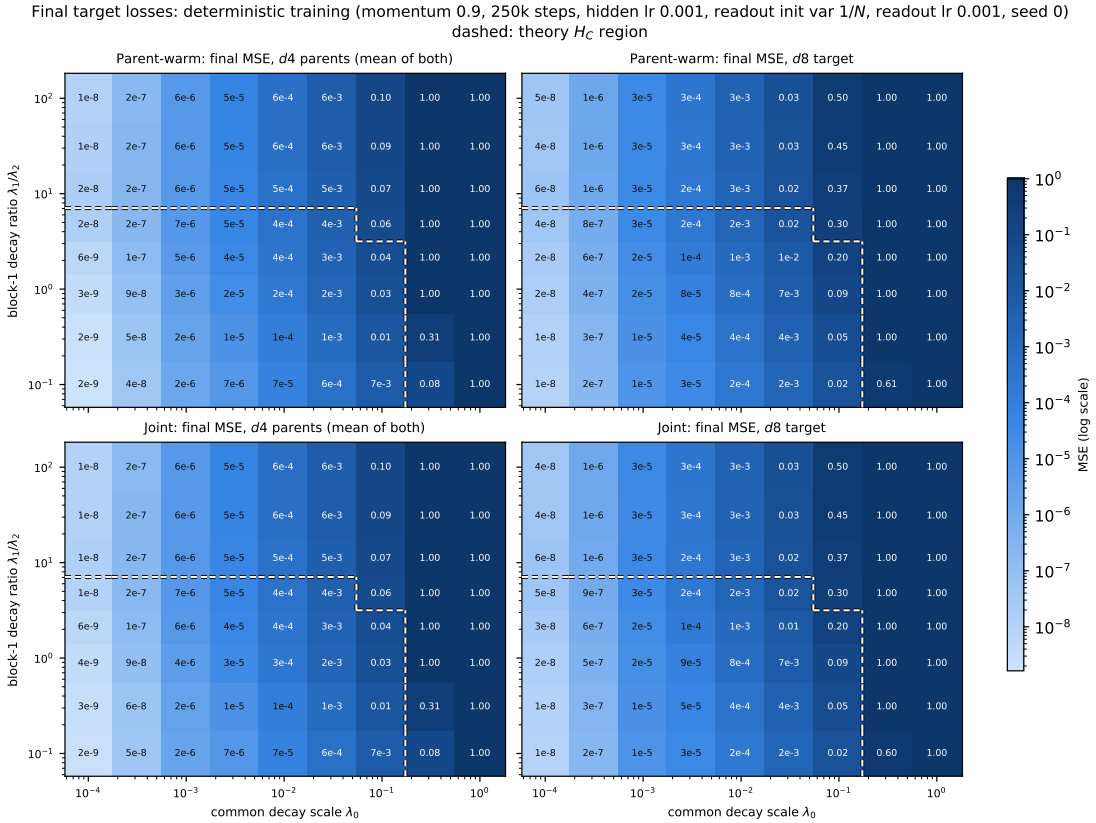


Figure 3: **Final losses:** MSE of the two $d4$ parents (mean) and of the $d8$ target, log scale.

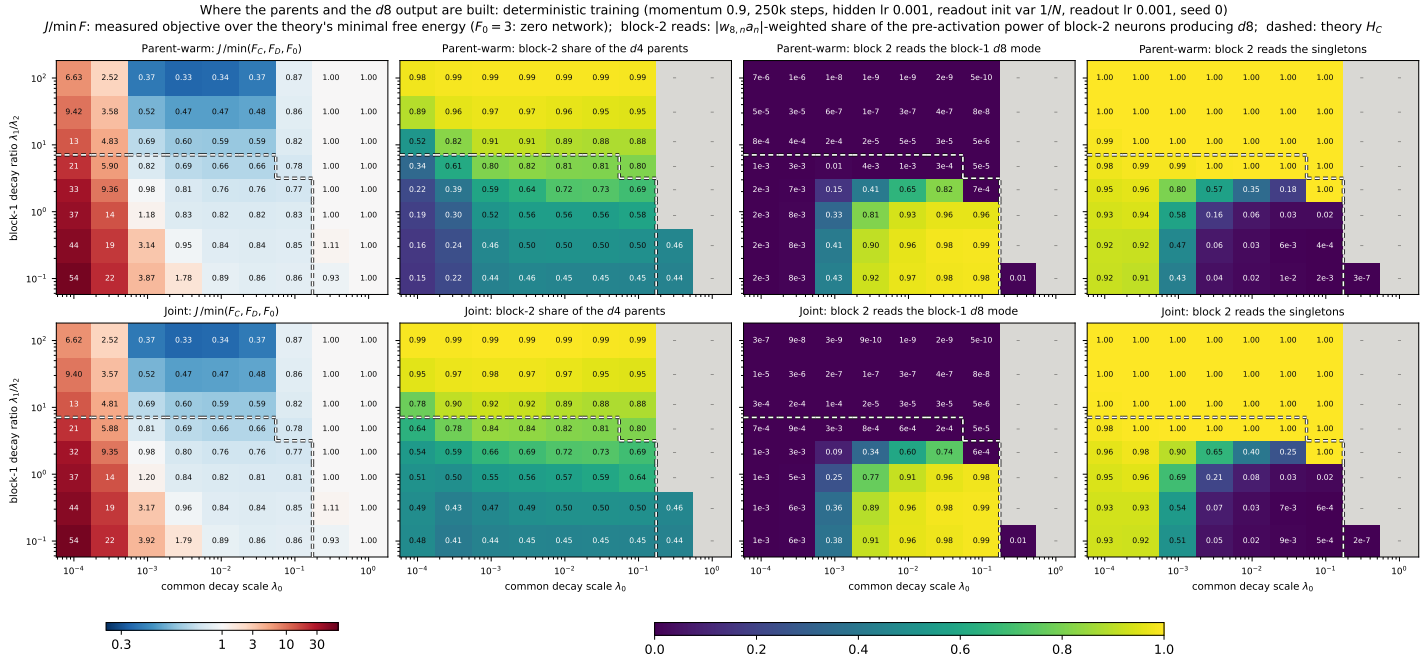


Figure 4: **Where the parents and the $d8$ output are built** (rows: parent-warm, joint). Columns: final objective over the theory's minimal free energy (blue: below it; red: above it, i.e. cells still converging); block-2 share of the readout-weighted parent amplitude; share of the pre-activation power of block-2 $d8$ producers (weighted by $|w_{8,n}a_n|$) on the χ_8 mode built in block 1 (relay of D_1) and on the eight singletons (direct D_2). Grey: block 2 produces no $d8$. The parent and degree-2 reads (≤ 0.09 outside $\lambda_0 = 0.3$) are not shown.